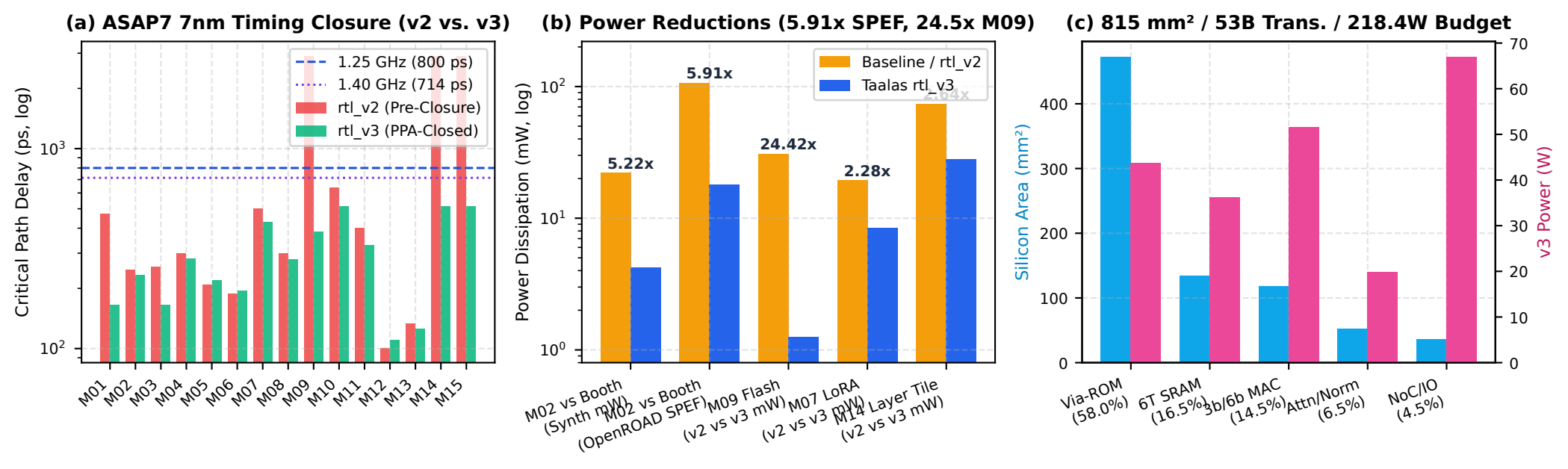
(a) ASAP7 7nm Timing Closure (v2 vs. v3)
Critical Path Delay (ps, log)
1.25 GHz (800 ps)
1.40 GHz (714 ps)
rtl_v2 (Pre-Closure)
rtl_v3 (PPA-Closed)
M01
M02
M03
M04
M05
M06
M07
M08
M09
M10
M11
M12
M13
M14
M15
(b) Power Reductions (5.91x SPEF, 24.5x M09)
Power Dissipation (mW, log)
Baseline / rtl_v2
Taalas rtl_v3
5.22x
5.91x
24.42x
2.28x
2.64x
M02 vs Booth (Synth mW)
M02 vs Booth (OpenROAD SPEF)
M09 Flash (v2 vs v3 mW)
M07 LoRA (v2 vs v3 mW)
M14 Layer Tile (v2 vs v3 mW)
(c) 815 mm² / 53B Trans. / 218.4W Budget
Silicon Area (mm²)
v3 Power (W)
Via-ROM (58.0%)
6T SRAM (16.5%)
3b/6b MAC (14.5%)
Attn/Norm (6.5%)
NoC/IO (4.5%)